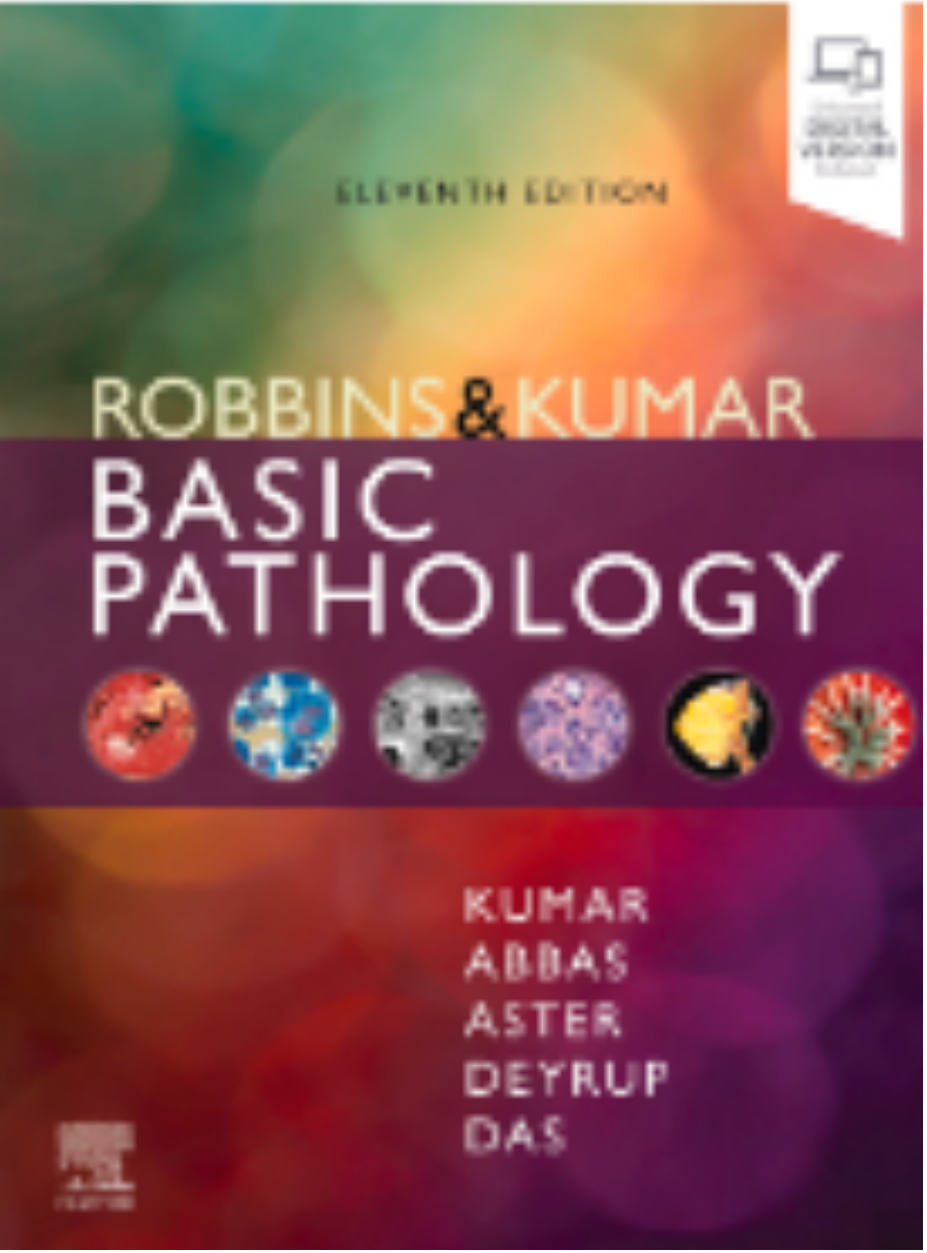


Diseases of Myelin

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Pathology Department


ELEVENTH EDITION
ROBBINS & KUMAR
BASIC PATHOLOGY
KUMAR
ABBAS
ASTER
DEYRUP
DAS


NO PHOTOGRAPHY
NO VIDEO RECORDING

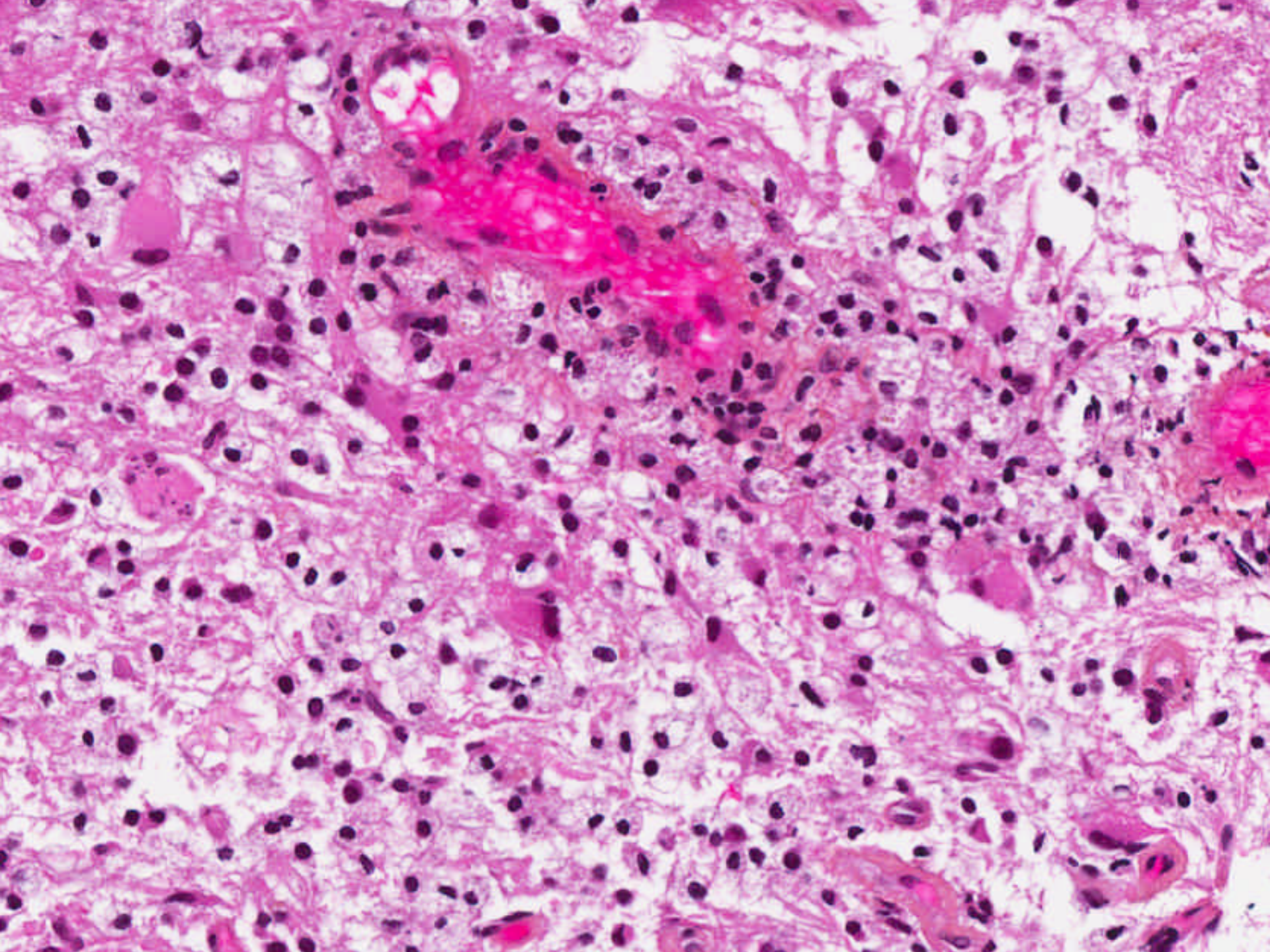
Learning Objectives

- Discuss multiple sclerosis in terms of pathogenesis, pathological features, and clinical features
- Describe the pathogenesis, pathology, and presentation of the most common demyelinating diseases of the central nervous system

Case

- A 30-year-old-woman presented to the ER with:
  - Seizure
- Past medical history:
  - Blurry vision in her left eye for 3 days before 2 months
    - Resolved spontaneously





Demyelinating Lesion
Diseases of Myelin

Diseases of Myelin

- Most diseases of myelin are white matter disorders
- Peripheral nerve myelin is similar to CNS myelin but differs in key aspects.
  - Made by Schwann cells, not oligodendrocytes
  - Each Schwann cell provides myelin for only one internode
  - The specialized proteins and lipids are different.
- Most diseases of CNS myelin do not involve the peripheral nerves to any significant extent, and vice versa.

Diseases of Myelin

- In general, CNS diseases involving myelin are separated into two broad groups:
  - Demyelinating diseases :
    - Acquired conditions characterized by damage to previously healthy myelin
      - MS, PML, drugs and toxin
  - Leukodystrophy, or dysmyelinating diseases:
    - Myelin is not formed properly or has abnormal turnover kinetics

Multiple Sclerosis

- An autoimmune demyelinating disorder characterized by episodes of disease activity, separated in time (dissemination in time), that produce white matter lesions that are separated in space (dissemination in space)
- It is the most common demyelinating disorder
- Any age, (childhood or >50 years of age is rare)
- Women are affected twice as often as men

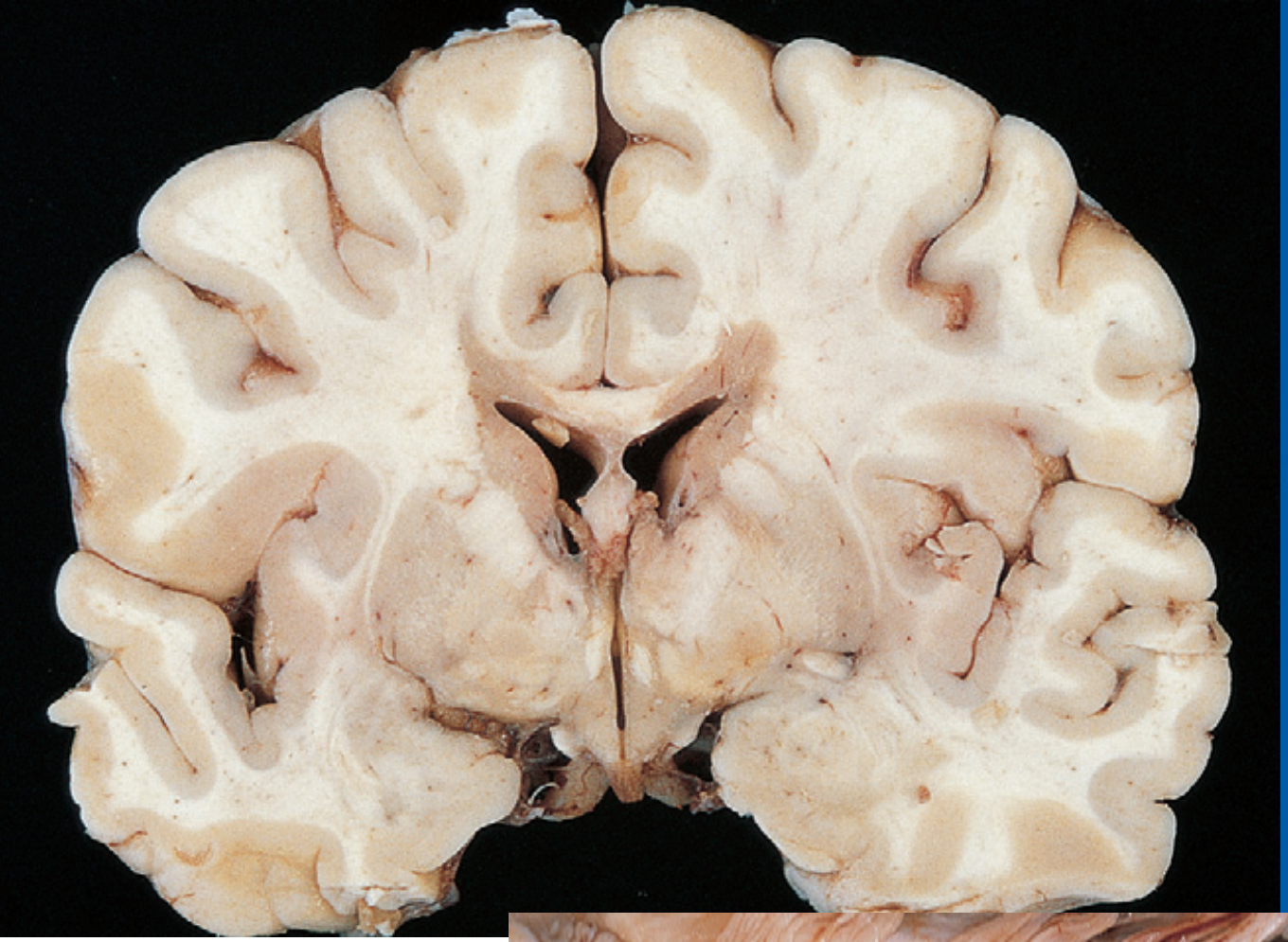
Pathogenesis

- Due to an autoimmune response directed against components of the myelin sheath in genetically predisposed individuals
- Closely related to genetic susceptibility and undefined environmental triggers
  - 15-fold higher incidence in first-degree relatives
  - ~150-fold higher risk in monozygotic twins
  - There is a strong effect of the major histocompatibility complex
    - Each copy of HLA-DRB1∗1501 allele increases MS risk ~3-fold
  - Other associated genes: IL-2 and IL-7 receptor genes



Autoimmunity

- Initiating cells: Th1 and Th17 T cells and B cells react against myelin antigens and secrete cytokines
- Th1 cells secrete IFN-γ → activates macrophages
- Th17 cells promote the recruitment of leukocytes
- Demyelination: caused by activated leukocytes and their damaging products
- Inflammatory infiltrate: mainly CD4+ T cells, some CD8+ T cells, and macrophages
- B cells and antibodies: contribute to disease, supported by the success of B cell–depleting therapies.
- EBV infection: associated with the disease, but its role in myelin autoimmunity remains unclear







Neuropathology: A Reference Text of CNS Pathology, 3e, David Ellison & Seth Love

2. الصورة الثانية — Myelin Staining

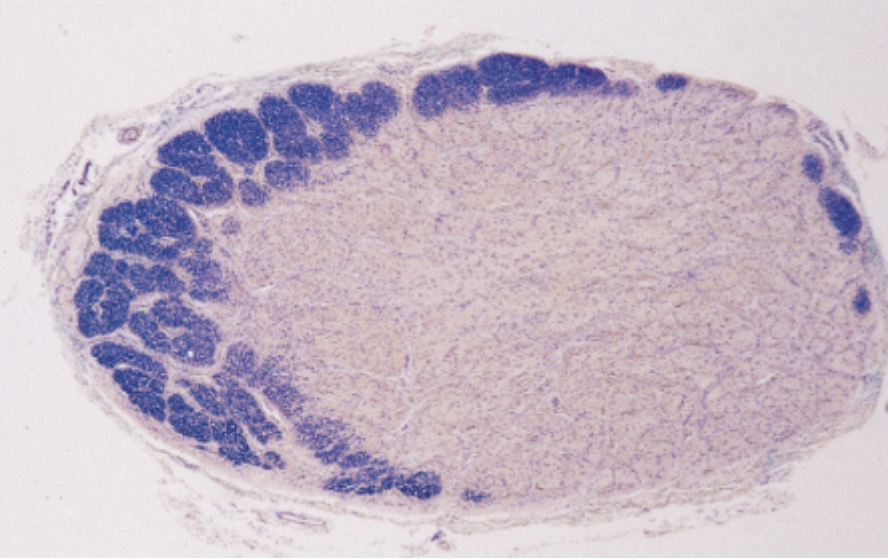
Luxol Fast Blue (LFB)

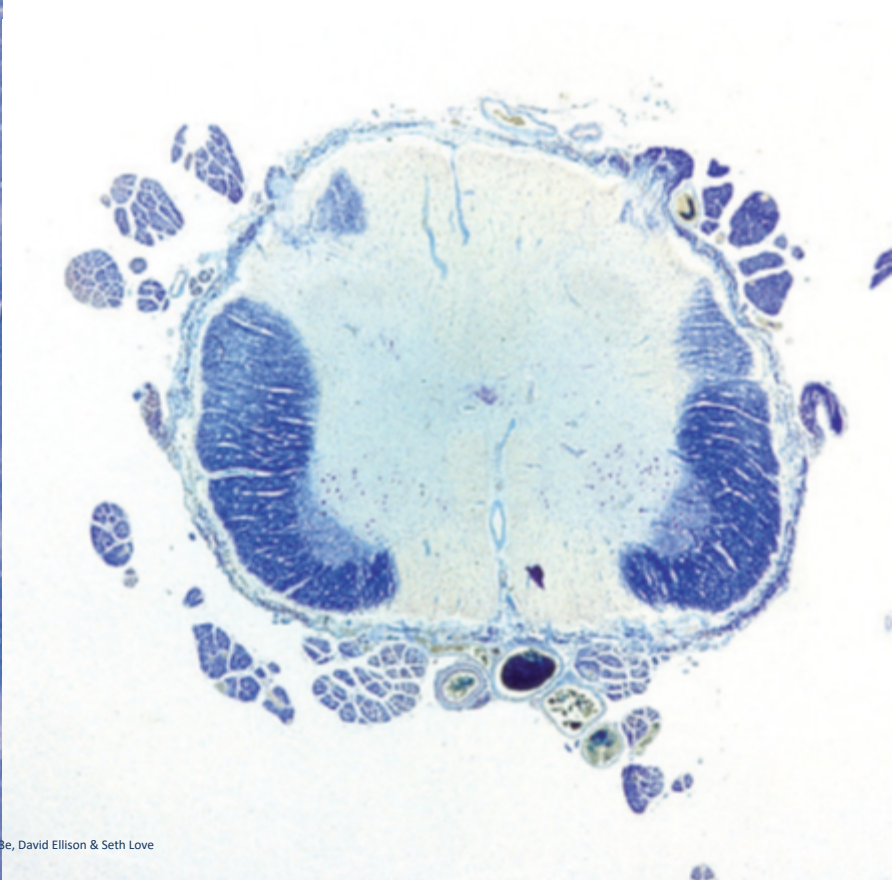
هذه الصبغة تُستخدم لإظهار المايلين باللون الأزرق.

- الأزرق الداكن: مناطق تحتوي على المايلين.
- **المناطق الفاتحة:** مناطق يقل فيها المايلين أو يكون مفقودًا.

في الصور الثلاث، لاحظ كيف تتجاور المناطق الزرقاء مع مناطق شاحبة واسعة، وهي آفات فقدان المايلين.







Neuropathology: A Reference Text of CNS Pathology, 3e, David Ellison & Seth Love



Kumar, Vinay. *Robbins & Kumar Basic Pathology*. 11th ed., Elsevier Enhanced Digital Version, 2022.

4. الصورة الرابعة — Normal vs Active Demyelination

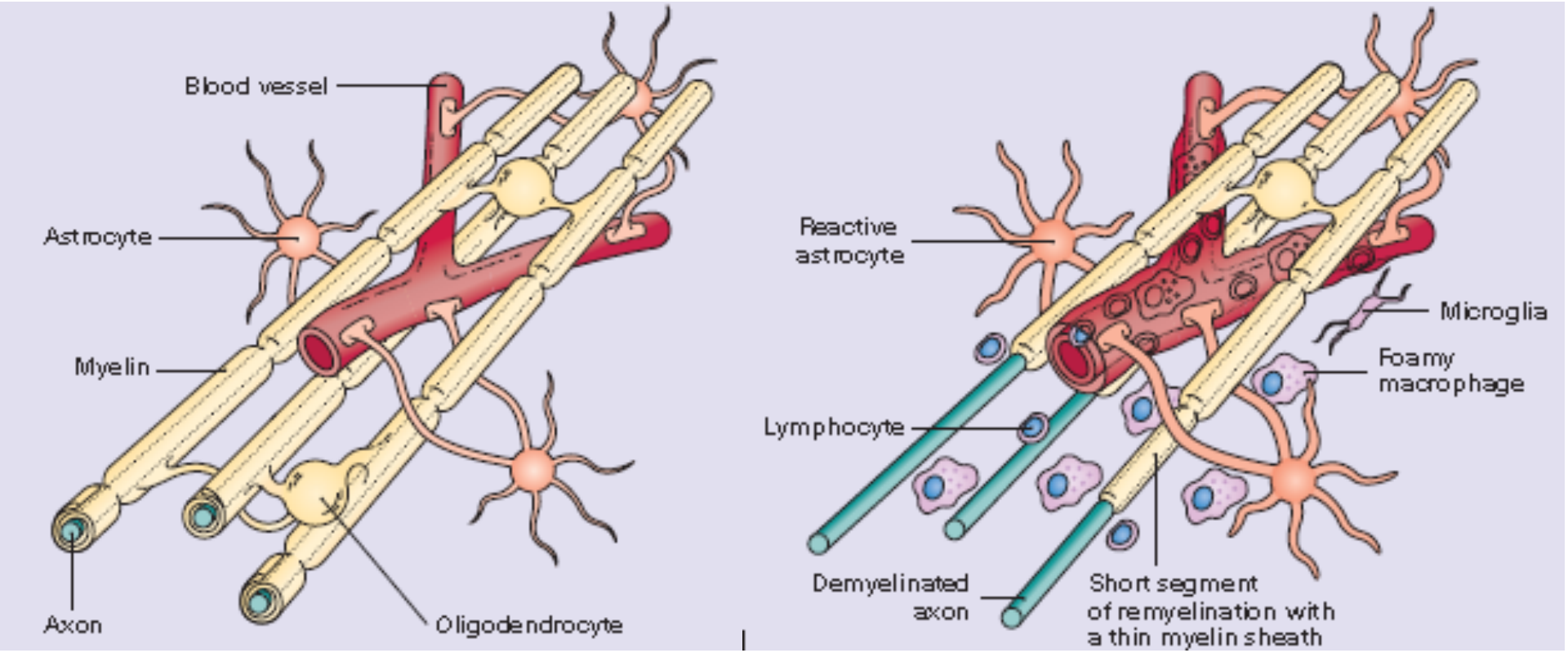
Normal (Left)

محاور عصبية محاطة بالمايلين الطبيعي، مع وجود الخلايا الدبقية والأوعية الدموية.

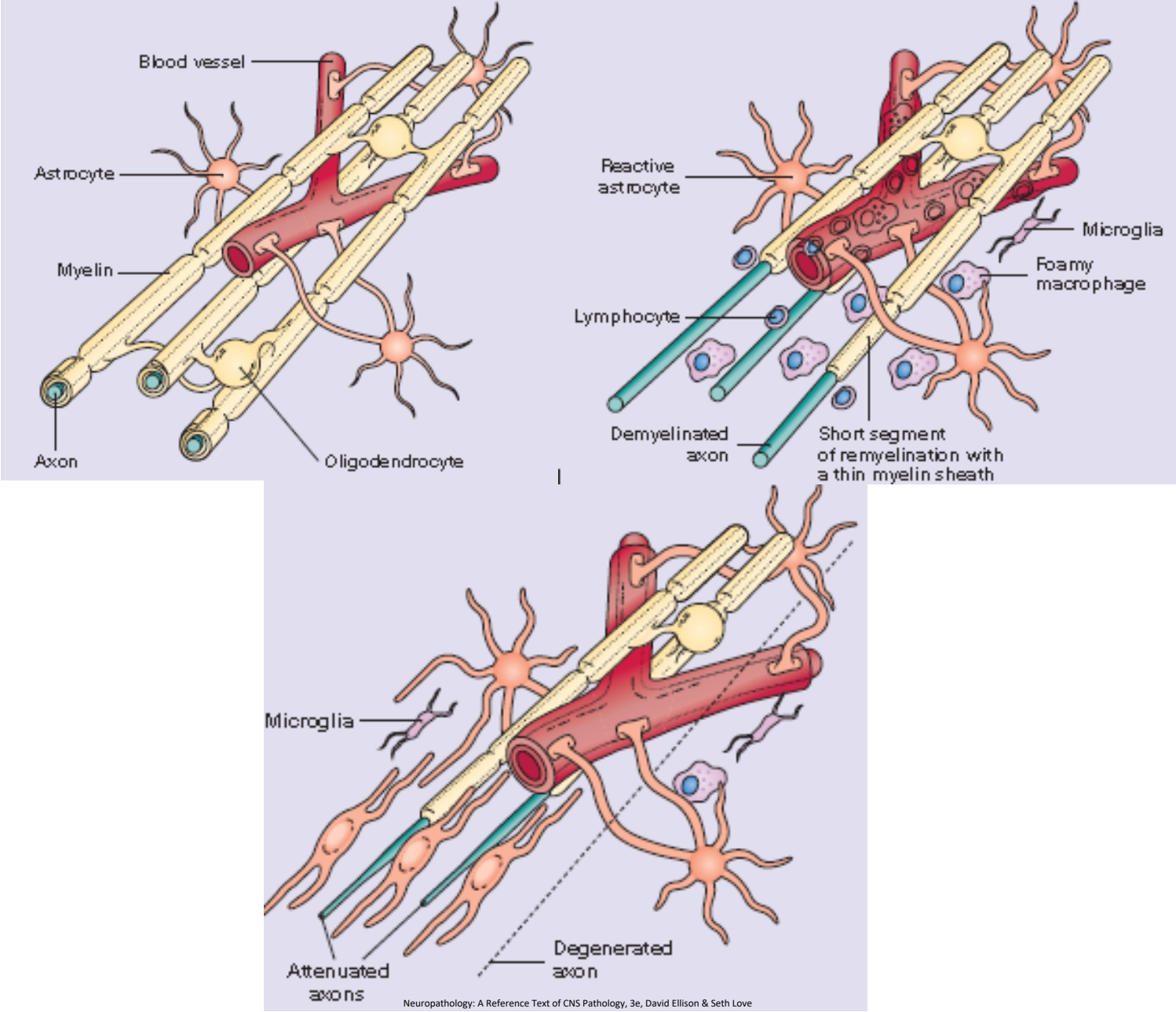
Active Lesion (Right)

فقدان المايلين، وتجمع خلايا التهابية وخلايا بلعمية رغوية، وظهور خلايا دبقية تفاعلية.

Foamy Macrophages: الخلايا البلعمية التي تبتلع بقايا المايلين المتضرر.



Neuropathology: A Reference Text of CNS Pathology, 3e, David Ellison & Seth Love



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Clinical Features

- The course of MS is variable:
  - Relapsing-remitting MS, up to 90% of cases
    - Characterized by multiple relapses and episodes of remission
    - Typically, recovery during remissions is not complete
    - Over time there is usually a gradual accumulation of neurologic deficits
  - Secondary progressive MS:
    - Has an initial relapsing-remitting phase then followed in 10 to 20 years by progressive worsening without remissions
  - Primary progressive:
    - Progressive disability from the outset, with little or no recovery

Clinical Features

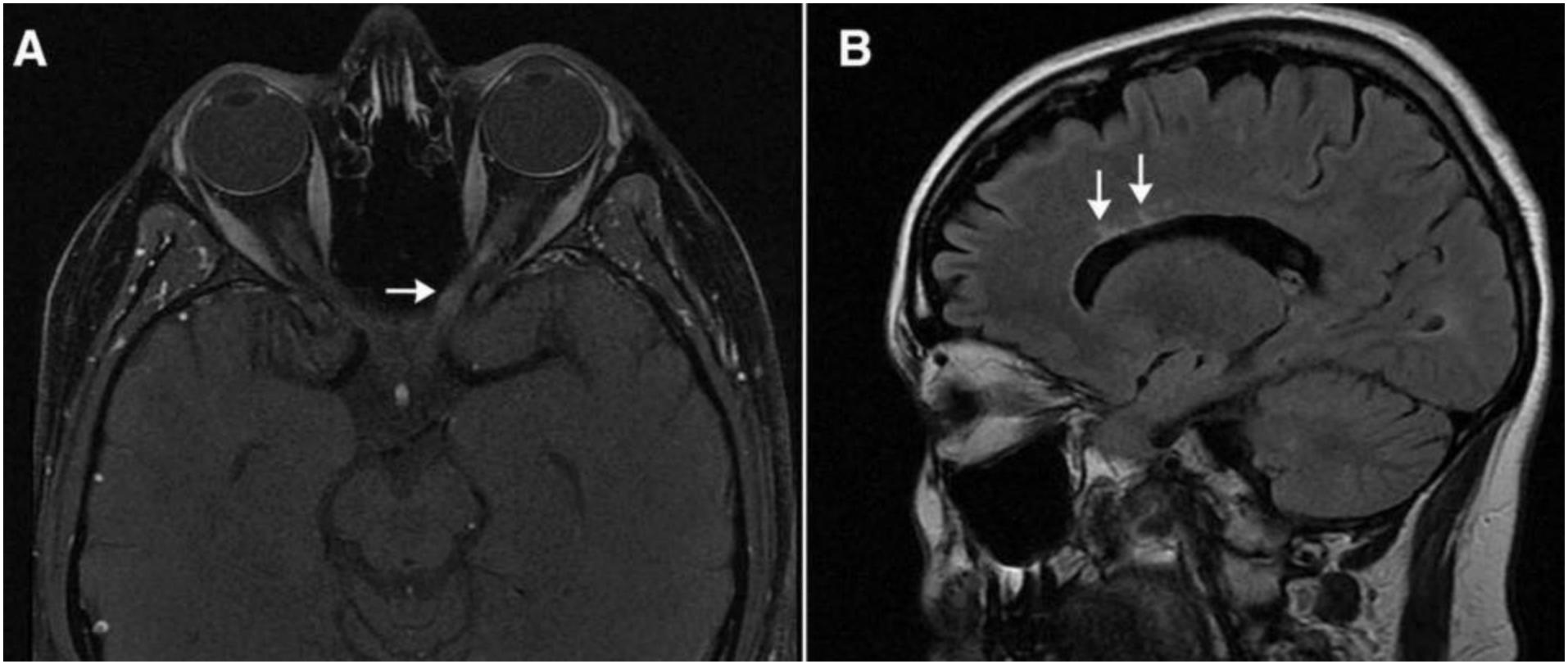
- It is difficult to predict when the next relapse will occur
  - Most current treatments (intended to control the immune response), aim at decreasing the rate and severity of relapses

Clinical Features

أول حدث عصبي سريري يوحي بوجود التهاب وفقدان للمايلين في الجهاز العصبي المركزي

- Clinically isolated syndrome (CIS):
  - Earliest clinical manifestations of MS
  - Unilateral visual impairment (optic neuritis)
    - Only 10% to 50% with an episode of optic neuritis go on to develop MS
- Symptoms, affected areas
  - Transverse myelitis: inflammation across the spinal cord parenchyma leading to neurologic dysfunction
    - Limb weakness, sensory loss, and bowel, bladder, or sexual dysfunction

Investigations



https://www.cmaj.ca/content/188/9/e199

Investigations

1. Lumbar Puncture (CSF Analysis):
   هذي الاجسام المضادة
   - Oligoclonal bands (OCBs): present in >90% of MS patients
   - ↑ IgG index
   - Mild lymphocytic pleocytosis
   - Mild ↑ protein
2. Evoked potentials: used to detect subclinical demyelination in other parts of the CNS.
   يشيك استجابة الجهاز العصبي لمحفز معين
   - Visual, somatosensory and brainstem auditory evoked potentials
3. Optical coherence tomograph

Investigations

ندور عن اي شي يدل على مرض ثاني

4. Blood Tests (to rule out mimics):
   - Aquaporin-4 (AQP4) antibody (neuromyelitis optica)
   - MOG antibody (MOG-associated disease)
   - ANA, anti-dsDNA, anti SM and others: SLE and vasculitis
   - Infections: HIV, syphilis, Lyme disease

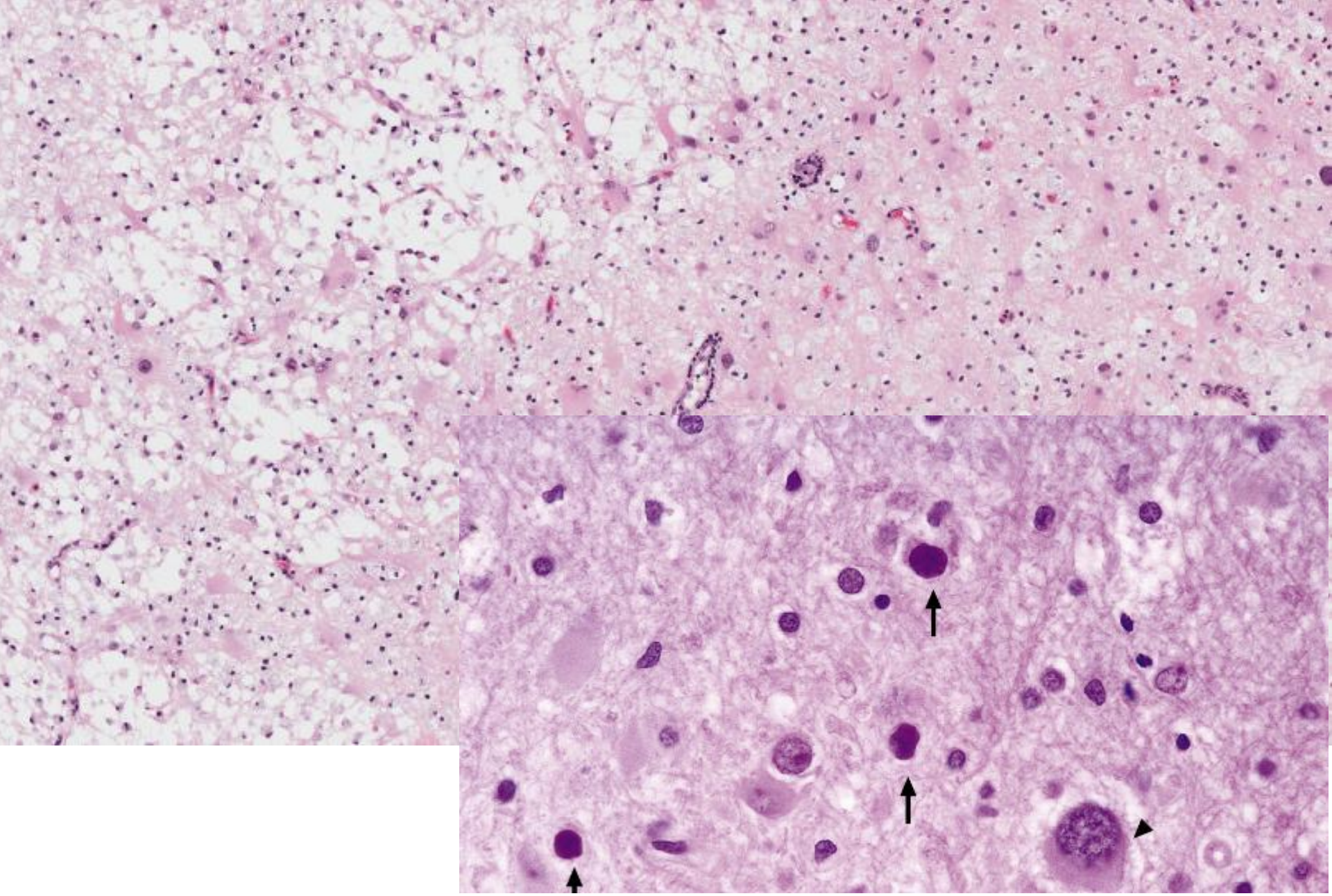
Diagnosis

- McDonald diagnostic criteria 2024:
  - Dissemination in space
  - Dissemination in time
  - CSF biomarkers



Case

- A-52-year-old women presented with:
  - Visual loss
  - Headache
- PMH:
  - AIDS x few years
- MRI:
  - Abnormal signal with nodular and meningeal enhancement



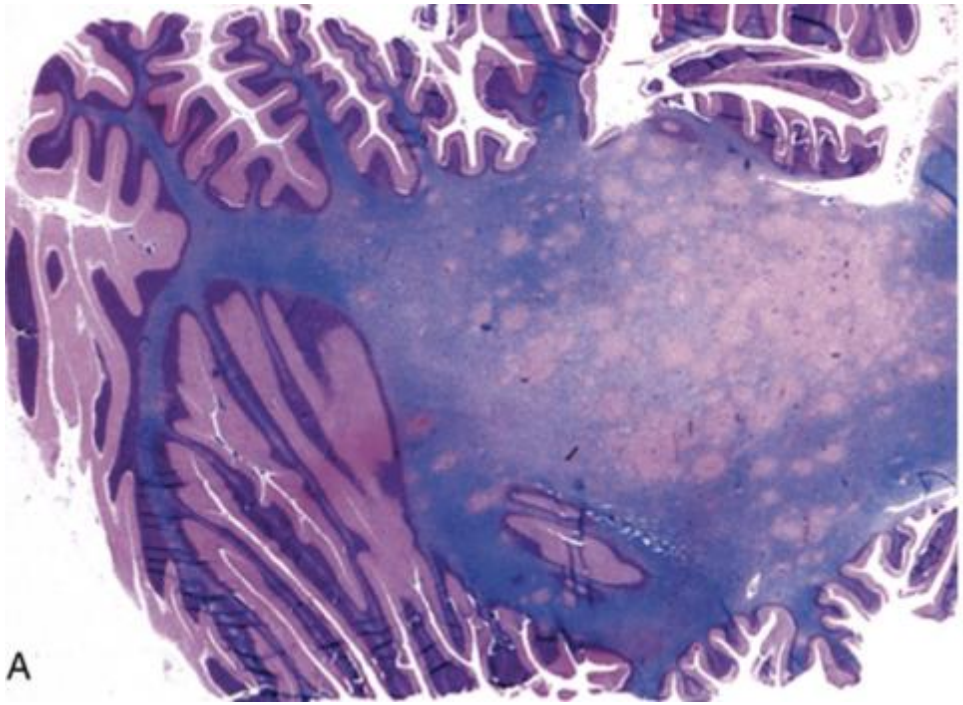
Pathology Department, Collage of Medicine, UQU, 2025-2026, Alaa Alkhotani©

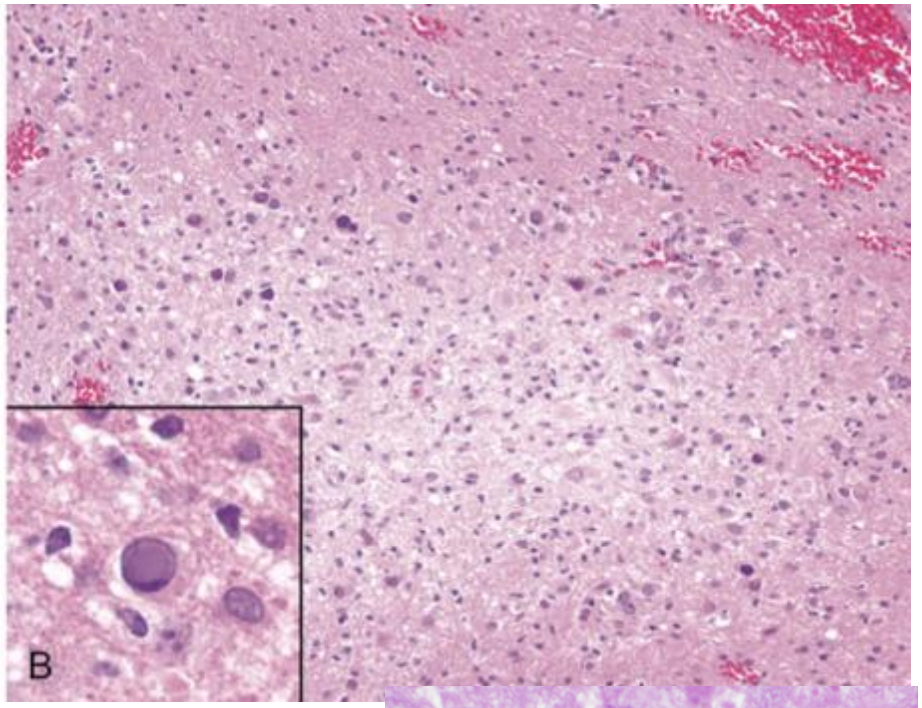
Progressive Multifocal Leukoencephalopathy

Progressive Multifocal Leukoencephalopathy

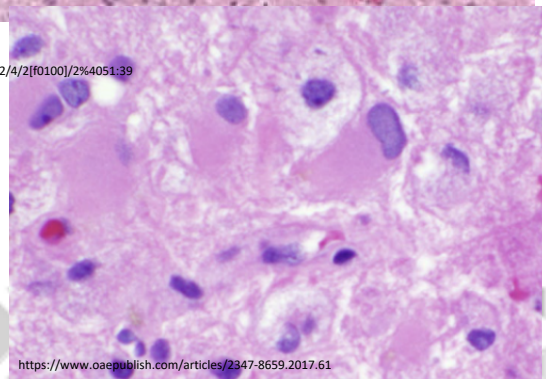
- Demyelinating disease that occurs after reactivation of the JC virus in patients who are immunocompromised.
  - Most people show serologic evidence of exposure to JC virus during childhood
  - JC virus, a polyomavirus, which preferentially infects oligodendrocytes
- Progressive neurologic signs and symptoms
- Imaging studies show extensive, often multifocal lesions in the hemispheric or cerebellar white matter
- No specific treatment , usually fatal in less than a year

Progressive Multifocal Leukoencephalopathy





https://bookshelf.health.elsevier.com/reader/books/9780323790215/epubcfi/6/734[%3Bvnd.vst.idref%3DCH0021_726-774_B9780323790185000217]!/4/2[c0021]/8/6/24/14/12/12/4/2[f0100]/2%4051:39



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